

JACUP, Australia

WMO: 956360

Lat: 33.886S

Lon: 119.111E

Elev: 1004

StdP: 14.17

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	36.9	38.4	27.9	22.6	68.3	31.4	26.6	60.1	25.8	56.7	24.0	56.9	6.9	0	0.586

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	22.5	94.7	65.0	89.5	64.2	85.0	63.8	70.5	81.6	67.8	79.9	65.9	78.0	13.3	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.6	105.4	73.9	64.7	94.9	69.6	62.7	88.5	67.5	35.0	80.3	32.7	80.3	31.2	77.8	97.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.3	21.1	19.1	33.3	104.9	1.5	3.8	32.2	107.6	31.4	109.8	30.5	111.9	29.5	114.7		
(5)			32.6	79.4	1.3	7.8	31.7	85.1	31.0	89.6	30.3	94.1	29.3	99.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.3	68.0	68.0	65.4	61.5	56.5	52.3	50.4	51.3	53.7	57.7	62.3	65.0		
	DBStd	8.27	6.14	5.82	6.56	5.76	4.75	3.72	3.62	4.08	5.08	6.03	6.59	6.07		
	HDD50	114	0	0	0	0	4	17	39	31	18	5	0	0		
	HDD65	2578	33	29	74	138	266	380	452	424	339	239	129	75		
	CDD50	3513	557	504	478	344	206	87	53	73	130	245	370	466		
	CDD65	501	125	113	86	32	4	0	0	0	2	15	48	76		
	CDH74	5314	1285	1032	788	295	40	1	0	6	54	262	646	905		
	CDH80	2258	599	451	344	90	6	0	0	0	11	88	272	397		
(14) Wind	WSAvg	10.6	12.2	11.8	10.6	9.3	9.4	10.1	10.4	10.0	10.3	10.4	11.3	11.5		
(15) Precipitation	PrecAvg	16.3	1.0	0.7	1.3	1.3	1.4	1.7	1.7	1.8	1.6	1.3	1.3	1.0		
	PrecMax	23.4	4.7	2.9	3.5	3.3	3.1	3.5	2.9	4.3	3.4	3.0	3.9	2.8		
	PrecMin	9.3	0.0	0.0	0.0	0.1	0.5	0.8	0.6	0.6	0.3	0.2	0.0	0.0		
	PrecStd	3.7	1.5	0.7	1.0	0.8	0.8	0.8	0.6	0.9	0.8	0.9	0.9	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	102.5	98.2	96.8	88.6	79.1	70.3	67.3	73.2	80.7	89.3	95.3	98.7		
		MCWB	69.3	66.1	64.7	63.7	59.5	54.3	52.9	56.7	60.6	61.9	63.1	63.8		
	2%	DB	93.0	91.6	89.0	81.7	73.5	66.2	63.6	67.0	73.4	81.3	87.9	89.7		
		MCWB	67.0	65.6	63.7	61.8	57.1	53.6	52.8	54.6	57.3	59.8	62.6	62.9		
	5%	DB	86.2	85.3	82.6	76.5	69.6	63.3	60.8	63.3	68.6	75.4	81.1	83.6		
		MCWB	66.1	65.4	62.9	60.4	56.2	53.0	52.1	54.0	55.5	58.5	61.0	62.9		
	10%	DB	80.5	79.8	76.8	72.1	66.1	60.4	58.4	60.3	64.7	70.4	75.6	77.9		
		MCWB	63.9	63.9	61.7	59.6	55.4	52.3	51.4	52.7	54.1	56.6	59.3	61.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.3	74.4	69.6	68.7	63.6	58.8	57.2	62.0	65.4	66.7	69.4	69.6		
		MCDB	87.1	83.4	79.7	75.5	70.8	64.1	61.3	63.8	69.4	77.4	82.5	84.8		
	2%	WB	70.4	69.4	67.1	64.9	60.4	56.5	55.1	57.7	59.2	61.9	64.5	65.9		
		MCDB	82.2	82.7	78.6	73.8	67.4	61.9	59.6	62.7	67.9	73.5	81.2	80.7		
	5%	WB	67.3	67.0	65.3	63.0	58.6	55.0	53.5	55.3	56.6	59.6	62.2	64.0		
		MCDB	80.7	78.1	75.7	71.0	65.2	60.0	58.6	61.1	65.4	71.6	75.7	78.0		
	10%	WB	65.1	65.3	63.7	61.1	57.0	53.5	52.0	53.5	54.8	57.6	60.3	62.2		
		MCDB	77.7	75.9	73.1	68.1	63.0	58.2	57.1	58.9	62.9	68.1	72.1	74.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	23.9	22.5	21.5	20.4	18.6	16.7	16.2	18.4	21.2	23.3	24.4	24.0		
		MCDBR	34.6	32.7	30.8	26.5	23.7	20.7	19.4	22.8	27.9	32.0	34.2	34.2		
		MCWBR	12.0	10.6	10.2	10.7	10.8	11.0	10.9	12.6	13.5	13.0	12.4	11.6		
	5% WB	MCDBR	29.2	27.2	23.9	21.6	18.2	17.0	16.4	19.3	23.8	26.9	29.6	29.6		
		MCWBR	13.4	11.1	9.6	11.6	10.3	10.8	10.5	12.6	13.8	12.9	12.3	11.9		
(40) Clear-Sky Solar Irradiance	taub		0.341	0.334	0.320	0.307	0.295	0.288	0.286	0.292	0.316	0.319	0.326	0.329		
	taud		2.530	2.562	2.592	2.624	2.623	2.618	2.618	2.594	2.502	2.525	2.532	2.544		
	Ebn at Noon		316	312	305	292	279	273	280	293	301	313	319	321		
	Edh at Noon		35	33	30	25	23	21	22	26	32	34	35	35		
(44) All-Sky Solar Radiation	RadAvg		2376	2084	1702	1262	946	794	853	1117	1493	1898	2207	2436		
	RadStd		167	135	105	77	61	36	43	44	53	106	138	146		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+189	N/A

Nomenclature: See separate page